

Trainings ▾

General Information

⚠ CAUTION ⚠

Extending the oil and filter change interval beyond the recommendations will decrease the engine life due to factors such as corrosion, deposits, and wear.

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A sulfated ash limit of 1.85 percent has been placed on all engine lubricating oils recommended for use in Cummins® engines. Higher ash oils can cause valve and/or piston damage and lead to excessive oil consumption.

The use of quality engine lubricating oils, combined with appropriate oil drain and filter change intervals, is a critical factor in maintaining engine performance and durability. Extending the oil and filter change interval beyond the recommendations will decrease engine life due to factors such as corrosion, deposits, and wear. Reference Procedure 102-002 in Section 2 to determine which oil drain interval to use for the application.

Cummins Inc. recommends the use of high-quality Society of Automotive Engineers (SAE) 15W-40 heavy-duty engine oil, such as Valvoline™ Premium Blue™ (USA) or Valvoline™ Premium Blue™ Extra (International).

Note : The responsibility is with the owner. If recommendations are ignored, warranty could be affected.

API: American Petroleum Institute

CES: Cummins® Engineering Standard

ACEA - Association des Constructeurs European d' Association

JAMA - Japanese Automobile Manufacturerrs Association

Table 1: CES for Lubricants

CES	North American Classification	International Classifications	Comments
Obsolete. Do not use.	API CD API CE	ACEA E-1	Obsolete. Do not use.
	API CG-4/SH		

Table 1: CES for Lubricants

CES	North American Classification	International Classifications	Comments
CES-20075 ¹	API CF-4/SG	ACEA E-2	Minimum acceptable oil classification for MidRange engines, but is not recommended.
		ACEA E-3	
		JAMA DH-1	
CES-20071 ²	API CH-4 4/SJ	ACEA E5	Acceptable oil classification for MidRange engines.
CES-20076 ²			
CES-20077 ²			
CES-20078	API CI-4	ACEA E7	Excellent oil for MidRange engines.
CES-20081	API1 CJ-4	ACEA E9	Excellent oil for MidRange engines where ultra-low sulfur diesel fuel is used. ³
		JAMA DH-2	

Table Notes

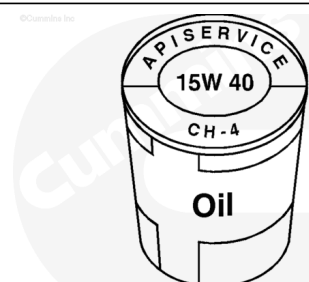
1. For MidRange engines, in areas where CH-4/SJ or CG-4/SH oils are **not** available, refer to the oil drain intervals in Section 2. As an alternative, oils meeting CES-20075 can be used, but the oil drain interval and filter change interval **must** be reduced by half.
2. Outside North America, where oil meeting CES-20071, CES-20076, or CES20077 might **not** be available, Cummins Inc. primary recommendation is for an oil meeting Global DHD-1, as jointly developed by EMA, ACEA, and JAMA.
3. Ultra-low sulfur diesel fuel is defined as diesel fuel **not** exceeding 0.0015 (15 ppm) mass percent sulfur content (ultra-low diesel fuel is also defined by ASTM S-15).

A sulfated ash limit of 1.0 mass percent is suggested for optimum valve and piston deposit and oil consumption control.

For further details and discussion of engine lubricating oils for Cummins® engines, refer to Fluids for Cummins® Products Service Manual, Bulletin 5411406.

The API service symbols are shown in the accompanying illustration. The upper half of the symbols display the appropriate oil categories.

The lower half can contain words to describe oil energy-conserving features.

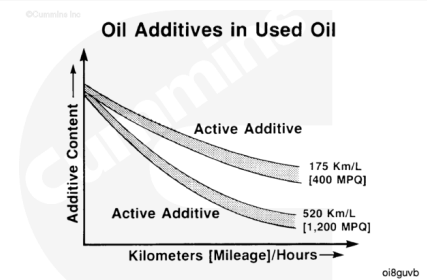


The center section identifies the SAE oil viscosity grade.

As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination of the oil between oil and filter change intervals is normal. The amount of contamination will vary, depending on the operation of the engine, kilometers or miles on the oil, fuel consumed, and new oil added.

Extending oil and filter change intervals beyond the recommendations will decrease engine life due to factors such as corrosion, deposits, and wear.

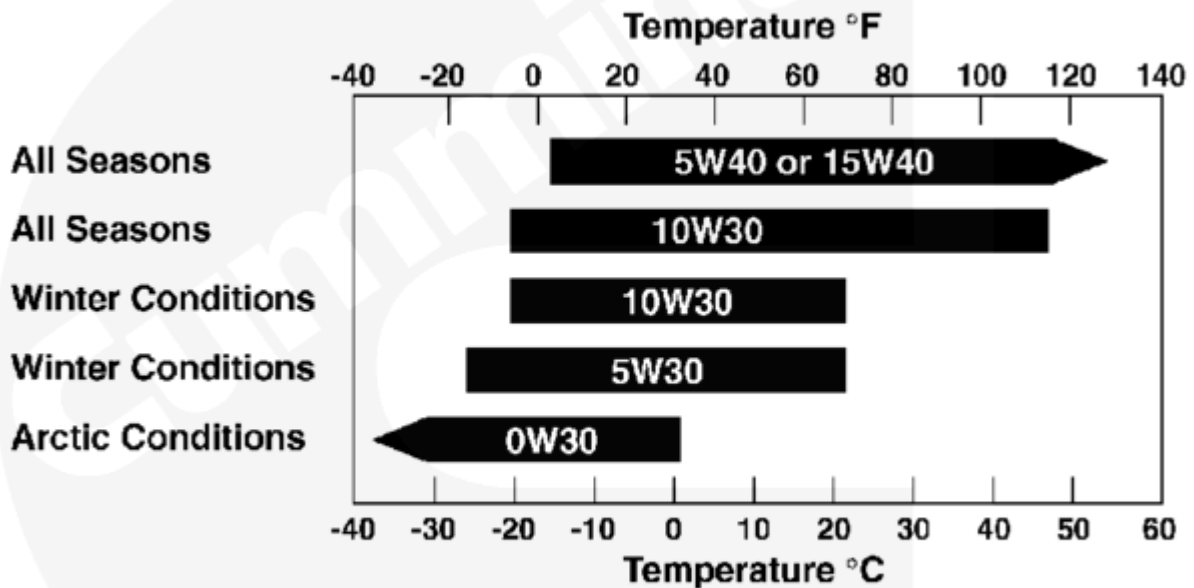
Reference the oil drain chart in this section to determine which oil drain interval to use for your application.



The primary Cummins Inc. recommendation is for the use of 15W-40 multigrade lubricating oil for normal operation at ambient temperatures above -15°C [5°F]. The use of multigrade oil reduces deposit formation, improves engine cranking in low temperature conditions, and increases engine durability by maintaining lubrication during high temperature operating conditions. Since multigrade oils have been shown to provide approximately 30 percent lower oil consumption compared with monograde oils, use multigrade oils to be certain the engine will meet applicable emissions requirements.

Use of "synthetic engine oils" (those made with API group 3 or group 4 base stocks) is permitted, subject to the same performance and viscosity limitations of petroleum (mineral) based engine oils. The same oil change intervals that are applied to petroleum (mineral) based engine oils **must** be applied to synthetic oils.

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While the preferred viscosity grade is 15W-40, lower viscosity multigrade oils can be used in colder climates. See the accompanying chart. Any viscosity grade lower than 15W-40 must still meet CES 20081.

Synthetic engine oils, API Group III and Group IV basestocks, are recommended for use in Cummins® engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Synthetic 0W-30 oils that meet the requirements of API Group III or Group IV basestocks, can be used in operations where the ambient temperature never exceeds 0°C [32°F].

Multiviscosity oils rated 0W-30 do **not** offer the same level of protection against fuel dilution as do higher multigrade oils. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

As these oils have directionally thinner oil films than 15W-40 oils, top-quality Fleetguard® filters **must** be used above 20°C [70°F]. Some oil suppliers might claim better fuel economy for these oils. Cummins Inc. can neither approve nor disapprove any product **not** manufactured by Cummins Inc. These claims are between the customer and the oil supplier. Obtain a commitment from the oil supplier that the oil will give satisfactory performance in Cummins® engines, or do **not** use the oil.

New Engine Break-in Oils

Special "break-in" engine lubricating oils are **not** recommended for new or rebuilt Cummins® engines. Use the same type of oil during the break-in as is used in normal operation.

AfterMarket Oil Additive Usage

Cummins Inc. does **not** recommend the use of aftertreatment oil additives. Present high-quality fully additive engine lubricating oils are very sophisticated, with precise amounts of additives blended into the lubricating oil to meet stringent requirements. These oils meet performance characteristics that conform to the lubricant industry standards. Aftermarket lubricating oil additives are **not** necessary to enhance engine oil performance, and in some cases, can reduce the finished oil's ability to protect the engine.